BEFORE YOU READ

Seven rules and three theorems

The page collects seven algebraic rules for computing derivatives and three theorems about derivatives on an interval.

- L0 **Derivative rules and theorems**
  - L1 **Algebraic rules**
    - L2 **Constant** $f' = 0$
    - L2 **Constant multiple** $(cf)' = cf'$
    - L2 **Power** nx^{n-1}
    - L2 **Sum and difference**
    - L2 **Product** $f'g + fg'$
    - L2 **Quotient** needs g ≠ 0
    - L2 **Chain** $f'(g) \cdot g'$
  - L1 **Interval theorems**
    - L2 **Mean Value Theorem**
    - L2 **Rolle's theorem** special case of the MVT with f(a) = f(b)
    - L2 **L'Hôpital's rule** 0/0 and ∞/∞ forms

TAKEAWAY

Rules compute, theorems reason

The rules turn derivatives into algebra; the theorems connect a derivative to the behaviour of the function across an interval.